



CS1002 – Programming Fundamentals

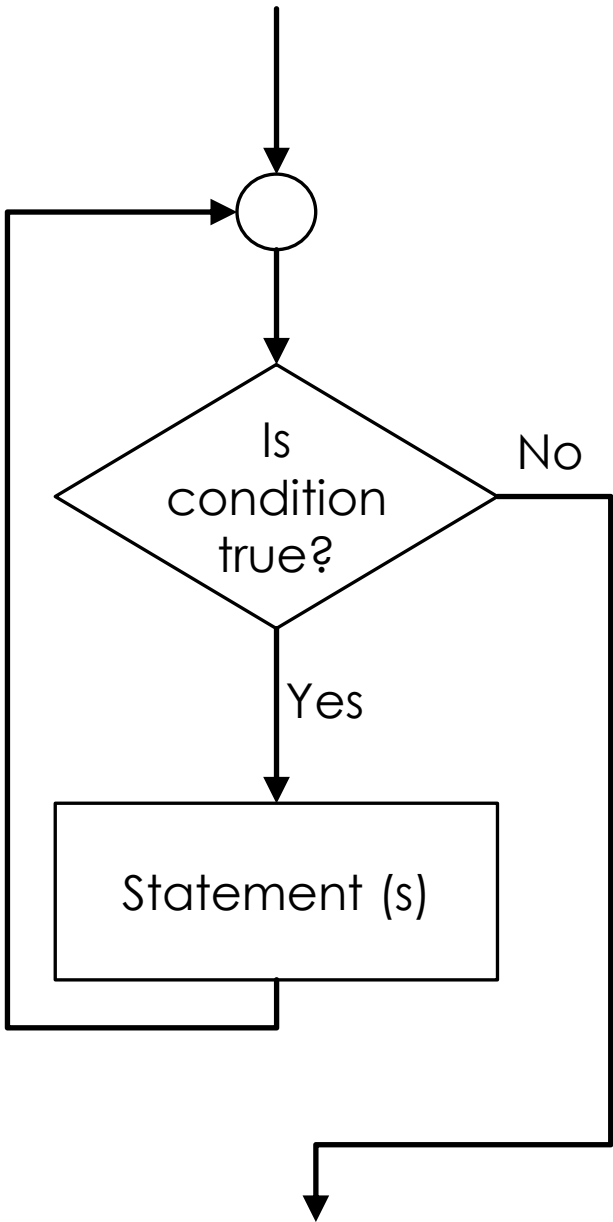
Lecture # 06
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FAST – NUCES, Faisalabad Campus

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Looping flowchart

While **condition is true** then
 Statement(s)
End while

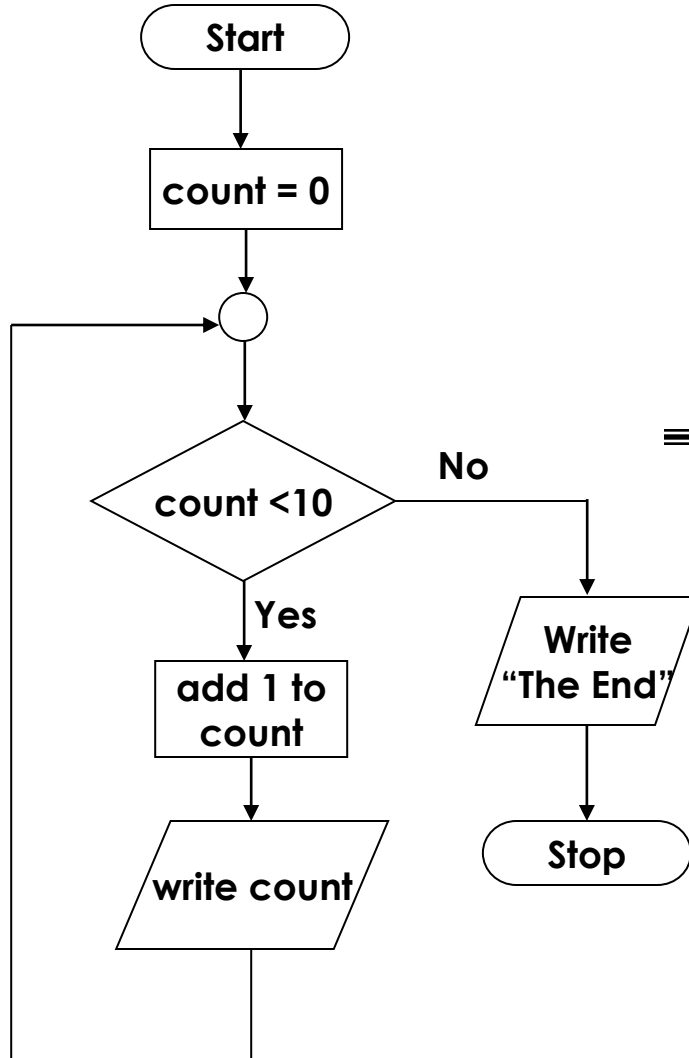


The Repetition Structure

- In flowcharting one of the more confusing things is to separate selection from looping
- This is because both structures use the diamond as their control symbol
- In pseudocode we avoid this by using specific keywords to designate looping

WHILE/ENDWHILE

WHILE / ENDWHILE



CS1002 - FALL 2021

```
count = 0
WHILE count < 10
    ADD 1 to count
    WRITE count
ENDWHILE
WRITE "The End"
```

Mainline

```
count = 0
WHILE count < 10
    DO Process
ENDWHILE
WRITE "The End"
```

Process

```
ADD 1 to count
WRITE count
```

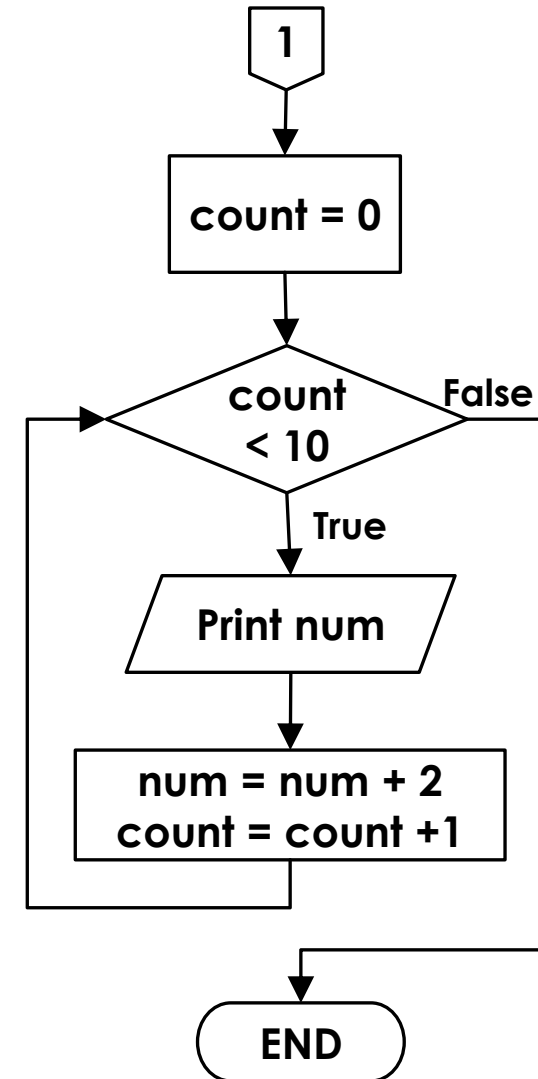
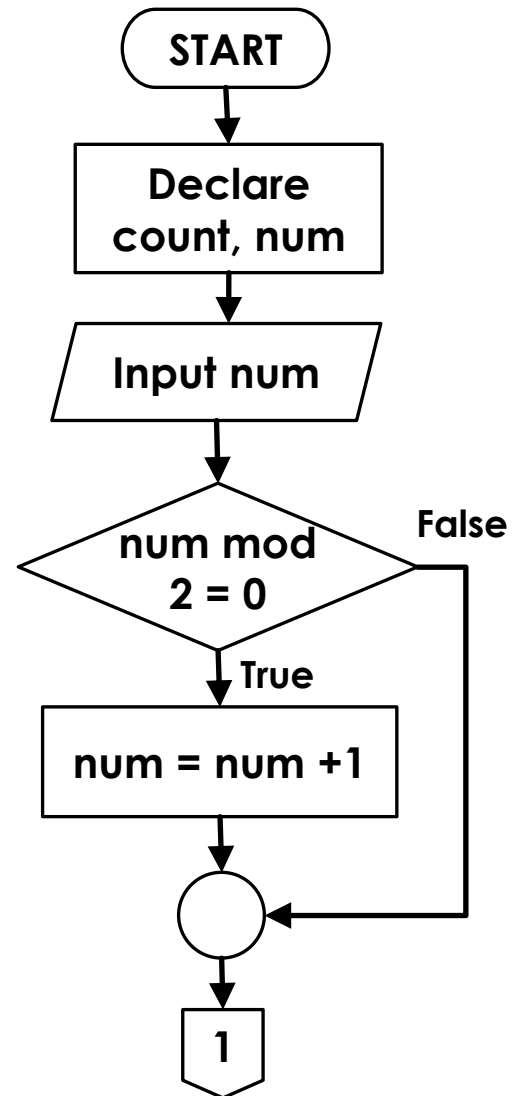
Example 2:

- Write pseudocode and draw a flowchart that will
- Print first 10 odd numbers starting from the number provided by the user.
- If it is even number then start from the next odd number if it is odd number then start with the current number
 - E.g. the user input 10 then the out put will be
 - 11 13 15 17 19 21 23 25 27 29
 - If the user input 5 then the output will be
 - 5 7 9 11 13 15 17 19 21 23
- **Hint:** Use $\text{number} \bmod 2 = 0$ to determine the even number

Pseudocode for printing 10 odd numbers

```
1.  START
2.  declare count, num
3.  input num
4.  if num mod 2 = 0 then
    4.1  num = num + 1
5.  end if
6.  count = 0
7.  while count < 10
    7.1  print num
    7.2  num = num + 2
    7.3  count = count +1
8.  end while
9.  END
```

Flow chart for printing 10 odd numbers



Example 3

- Write a pseudocode and draw a flowchart to
- Read an employee number (EMPNO), employee name (NAME), overtime hours worked (OVERTIME), hours absent (ABSENT) and
- Determine the bonus payment (PAYMENT) for 10 employees one by one

Bonus Schedule	
OVERTIME – $(2/3)*ABSENT$	Bonus Paid
>40 hours	5000
>30 but $\leq$ 40 hours	4000
>20 but $\leq$ 30 hours	3000
>10 but $\leq$ 20 hours	2000
$\leq$ 10 hours	1000

Example 4

- Write a pseudocode and draw a flowchart
- that will get 10 numbers from the user and print their average.
- Use only one number as input i.e. n1
- **Hint:** Use loop to get input in n1 and add it to Sum

Exercise Program 5: Square/cube

- Write a pseudocode and draw a flowchart for a program that will
 - Get one number from user
 - Print its square if it is an even number
 - Print its cube if it is an odd number
- The loop will run 5 times

Questions

